



FOREST VERIFICATION REPORT

ANALYZED ZONE INFORMATION

Latitude	40.9727	Longitude	39.7297
Zone size (km)	10	Analysis date	2026-09-29

47

FOREST INTEGRITY SCORE / 100 Degraded Forest

DETAILED RESULTS

Indicator	Value	Interpretation
Current vegetation (NDVI)	0.47	Good
NDVI Trend 3 years (Sentinel-2)	-0.009	Stable
10-year trend (Landsat 8/9)	+0.236	Improving
40-year trend (Landsat 1984)	+0.232	Improving
Forest Status (FAO)	Forest recovery	Recuperation
Annual NDVI rate	+0.58% / year	Amelioration
NDVI change 5 years	-0.004	Stable
Leaf type	Broadleaf (Deciduous/Evergreen)	Flat leaves, high photosynthesis
Developmental stage	Growing forest (20-60 years)	Active growth phase
Above-ground biomass	~36 t/ha	Low biomass (<50 t/ha)
Canopy Height (GEDI+S2 10m)	6.4m max / 2.3m median	Tree height from LiDAR (ISS) fused with Sentinel-2 at 10m resolution
Carbon Assessment (IPCC)	614,183 t CO2	CO2 sequestered in this forest zone — IPCC methodology (AGB+BGB × 0.47 × 3.67)
CO2 per hectare	78.2 t CO2/ha	Carbon density per hectare
Zone area	7,854 ha	Analysis zone area in hectares
Carbon credit value	EUR 9,212,745	Estimated value at EUR 15/tonne CO2 (voluntary carbon market)
REDD+ MRV Monitoring	Compliant	This report provides satellite-based Measurement, Reporting and Verification (MRV) data for the analyzed forest zone, consistent with REDD+ monitoring requirements. CO2 sequestration calculated using IPCC Tier 1 methodology from NASA GEDI LiDAR biomass data.

Methodology	IPCC 2006/2019 — Tier 1	Above-ground biomass measured via NASA GEDI LiDAR (25m, ISS). Carbon fraction: 0.47. CO2 conversion factor: 3.67. Below-ground biomass ratio: 0.26 (IPCC root-to-shoot). Source: IPCC Guidelines for National GHG Inventories (2019).
Carbon credit potential	EUR 767,728 / year	Estimated annual value based on maintained forest preservation. For official REDD+ credit issuance under VCS or Gold Standard, independent third-party verification is required.
GEDi Tree Cover	19.7%	Percentage of the zone covered by trees according to NASA GEDI
AI Forest Type	Boreal/Coniferous Forest	Coniferous, cold climate
Sentinel-1 Radar (VV)	-9.599 dB	Dense forest (radar)
Forest cover	67.3%	Good
Fire pixels detected (FIRMS/year)	102 pixels	~26 events estimated
Overall trend	stable	
Satellites used	- ESA Sentinel-2 - NASA FIRMS - ESA Sentinel-1 - NASA Landsat 8/9 - NASA Landsat 4/5 - NASA GEDI LiDAR	
Deforestation Alert	STABLE — No significant change	
Area change (3 years)	-3 ha	-0.03%
Annual rate (area/yr)	-1 ha/yr	Loss
Area change (10 years)	+8813.8 ha	+10369.18%
Annual rate 10yr (area)	+881.4 ha/yr	Recovery

SCORE BREAKDOWN

Component	Points	Maximum
Current vegetation (NDVI)	11	25
NDVI trend (3 years)	8	25
Forest cover	18	25
Sentinel-1 Radar (VV)	10	15
Fire score (0=many fires, 10=none)	0	10
TOTAL	47	100

SPECIES CLIMATE SUITABILITY (2050)

Scenario: SSP2-4.5 (2050) — 5-model ensemble | Temperature: 13.3°C → 14.7°C (+1.5°C) | Precipitation: 861 → 577 mm/year (-33%)

Species	Family	Score 2050	Status	Temp. range
Vinca major	Apocynaceae	0.7	Suitable	-5.0–19.0°C
Neotinea tridentata	Orchidaceae	0.54	Moderate risk	0.3–12.5°C
Cephalanthera longifolia	Orchidaceae	0.4	Moderate risk	-1.1–12.5°C
Orchis simia	Orchidaceae	0.28	At risk	0.6–11.9°C
Ophrys scolopax	Orchidaceae	0.2	At risk	2.1–11.0°C
Muscari armeniacum	Asparagaceae	0.13	At risk	-4.1–10.9°C
Tussilago farfara	Asteraceae	0.05	At risk	-4.6–10.4°C
Primula vulgaris	Primulaceae	0	At risk	-3.6–7.7°C
Staphylea pinnata	Staphyleaceae	0	At risk	-2.3–5.9°C
Veronica peduncularis	Plantaginaceae	0	At risk	-2.0–7.8°C
Veronica filiformis	Plantaginaceae	0	At risk	-6.1–8.0°C
Athyrium filix-femina	Athyriaceae	0	At risk	-7.8–10.0°C
Digitalis ferruginea	Plantaginaceae	0	At risk	-1.5–9.8°C
Teucrium chamaedrys	Lamiaceae	0	At risk	-2.3–8.8°C
Asplenium scolopendrium	Aspleniaceae	0	At risk	-4.9–7.3°C

METHODOLOGY — Bioclimatic Envelope Assessment (BEA): Suitability scores are calculated by comparing projected 2050 temperature (NASA GDDP-CMIP6, 5-model ensemble: ACCESS-CM2, MIROC6, MPI-ESM1-2-HR, GFDL-ESM4, BCC-CSM2-MR, SSP2-4.5 scenario) with each species real thermal tolerance range, derived from global distribution analysis via GBIF (100 occurrences/species). Method: temperature estimated from latitude ($27 - |\text{lat}| \times 0.52^\circ\text{C}$), proportional penalties when 2050 temperature exceeds tolerated range. Thresholds: Suitable ≥ 0.70 | Moderate risk 0.40-0.69 | At risk < 0.40 . This BEA approach is consistent with full MaxEnt results for general trends (cf. published studies on Ceiba pentandra, SSP2-4.5, Frontiers in Forests 2026). Data sources: GBIF (CC BY 4.0) · WorldClim V1 · NASA GDDP-CMIP6 · ESA Sentinel-2.

FIRE DETECTION METHODOLOGY: Fire pixels are detected by NASA FIRMS using VIIRS (375m) and MODIS (1km) thermal anomaly sensors. Raw pixel count represents thermal detections over 12 months in the analysis zone. Estimated fire events are calculated using NASA clustering methodology (~4 pixels per fire event average). This is a statistical approximation. For precise fire event mapping, consult NASA FIRMS directly at firms.modaps.eosdis.nasa.gov